# Toward Clinical Applications of Intelligent Robotic Ultrasound Systems

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***Abstract*—The Robotic Ultrasound System (RUSS) has the potential to transform medical imaging by addressing limitations such as operator dependency, diagnostic variability, and reproducibility in traditional ultrasound (US) examination. Despite rapid technological advancements, a substantial gap remains between RUSS research progress and clinical adoption. This review examined the clinical roles and engineering advances of RUSS, identifying key barriers to translation. Clinically, it evaluated the current applications of RUSS in supporting US procedures, while from an engineering standpoint, it summarized recent innovations and remaining technical challenges. This review examined the current state-of-the-art RUSS technologies, categorizing them based on diverse organ-specific applications while also analyzing their core functional capabilities. This review revealed a focus disparity: while abdominal US is the most commonly used in clinical practice, vascular-targeted RUSS dominates current research. It also highlighted a misalignment between research priorities and actual clinical tasks. Current studies predominantly focused on autonomous scanning and imaging, with limited attention to downstream tasks such as disease diagnosis and analysis. Building on these observations, it identified critical challenges and future trends in RUSS development. This work provides a foundation for future research, fostering collaboration between clinicians and engineers to accelerate the translation of next-generation RUSS from bench to bedside.**

***Index Terms*—Ultrasound imaging, robotic ultrasound systems, medical robotics, clinical applications, autonomy, robot learning.**

## I. Introduction

ULTRASOUND (US) imaging techniques have been widely applied across clinical fields due to their non-invasive nature, low cost, absence of radiation, portability, and capacity to display real-time organ motion [1]. Its extensive application is evident from the 8.37 million US examinations conducted in England between November 2019 and November 2020, as shown in Fig. 1 —a figure notably higher than the 5.48 million Computed Tomography (CT) scans and 3.07 million Magnetic resonance imaging (MRI) scans performed during the same period [2]. Reflecting its growing demand, the global US equipment market was valued at $9.86 billion in 2024 and is projected to reach $22.09 billion by 2034, with a compound annual growth rate (CAGR) of 8.4% from 2024 to 2034 [3].

Despite its widespread use and significant market growth, several challenges have limited the full realization of US imaging's potential in expanding its role in disease diagnosis and treatment guidance [4]. Clinical US scanning relies on manual probe control, where the operators' expertise is crucial due to the complexity of interpreting US images and the limited field of view. Variability among operators can lead to inconsistencies in imaging quality and reproducibility [5], while the reliance on skilled sonographers restricts the accessibility of US services and imposes a high cognitive and technical burden. Moreover, Work-Related Musculoskeletal Disorders (WRMSDs) are a significant concern in sonography, with 91% of sonographers reporting related symptoms [6], [7], as shown in Fig. 1. This

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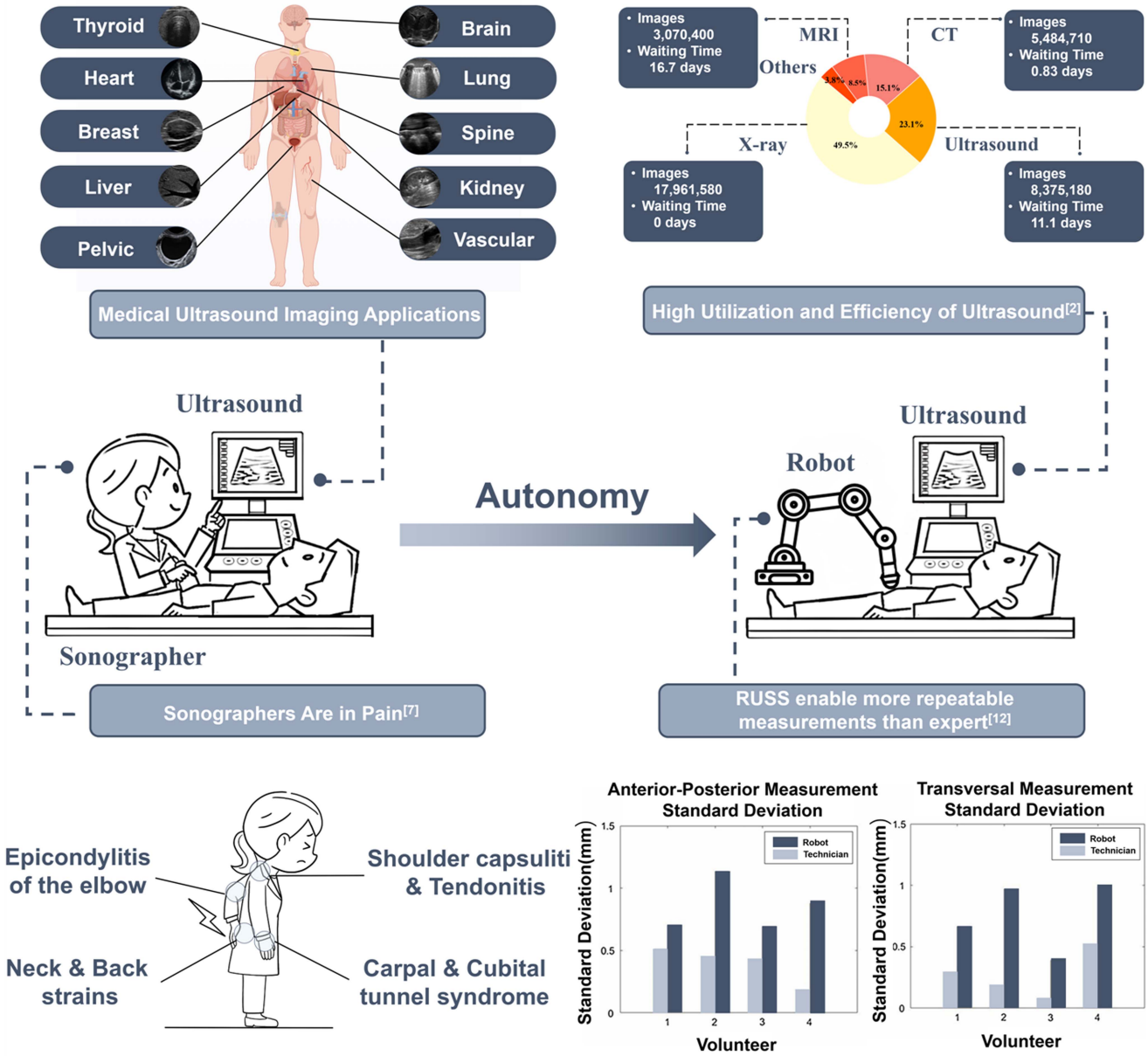



Fig. 1. Overview of US imaging and RUSS: The transition from traditional manual US systems to RUSS is primarily driven by the advancement of autonomy.

high prevalence not only impacts the well-being of healthcare professionals but also imposes a financial burden [8], [9]. To address these challenges, advancements in technology have explored the integration of robotics into US imaging. By combining the precision and automation of robotics with the real-time capabilities of US, The Robotic Ultrasound System (RUSS) aims to overcome many of the limitations inherent in traditional US practices [10].

In certain specialized fields, robots can complement human capabilities with their ability to execute complex and repeatable probe operations. With their potential for high precision, dexterity, and repeatability, robots are often suited for US integration [11]. They are particularly effective for repetitive or intensive tasks and can help reduce human error. Notably, Risto et al. [12] have shown that RUSS achieves higher measurement repeatability compared to expert-performed US imaging, as illustrated in the bottom-right panel of Fig. 1. RUSS combines robotic systems with US technology and represents a distinct category within medical imaging robotics [13], as shown in Fig. 1. Over the past few decades, several RUSS have been developed, with researchers exploring different technical approaches [14]. However, the development of RUSS has lagged behind advancements in other medical robotic applications. RUSS is not fully mature and is far from being widely adopted in clinical practice. For instance, the Da Vinci Surgical System (DVSS) is an advanced laparoscopic technology [15]. By the end of 2022, more than 7500 DVSS will have been installed worldwide, resulting in more than 12 million minimally invasive surgeries [16].

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The "valley of death" describes the gap between scientific research conducted in laboratories and its translation into clinical applications or commercialization [17]. Many ideas developed in research laboratories fail to make it to the marketplace. In the field of medical robotics, a "valley of death" phenomenon poses significant innovation challenges. This gap arises from limited communication and collaboration between engineers and clinicians, leading to a disconnect between technological advancements and their effective clinical translation or commercialization [18], [19]. Despite breakthroughs in robotic technologies, many innovations fail to progress beyond laboratory prototypes to become clinically adopted solutions. This lack of integration not only hinders the industrialization of research outcomes but also impedes the broader impact of medical robotics on healthcare. Focusing specifically on the application of RUSS, Jiang et al. [20] examined the annual number of peer-reviewed publications and the cumulative total over the years. The number of publications has grown exponentially, reaching an impressive total of 125110 between 2001 and 2022 [20]. However, the number of RUSS that is used on a large scale in medical clinics is limited. Developing a RUSS capable of performing clinical examinations in real-world settings requires the collaboration of sonographers, robotics engineers, entrepreneurs, and other stakeholders.

This review aims to clarify the gap between scientific research and real-world clinical applications in RUSS. It will summarize and analyze the gap from the perspectives of different clinical tasks and target organs. In addition, we will summarize the problems faced by RUSS in the clinic and how existing technologies are attempting to overcome these problems. This review aims to inspire clinicians and engineers to integrate innovative methods into the development of intelligent RUSS from two perspectives:

1) From a clinical perspective: we explored the current capabilities of RUSS in supporting clinical US procedures, highlighting the potential benefits it offers. Furthermore, we identified the key barriers hindering its widespread clinical adoption and discussed the steps needed to overcome these challenges
2) From an engineering perspective: we explored the current state of the art in the application of RUSS. While increasing autonomy has led to the development of more intelligent RUSS, several challenges remain unresolved.

To show the distinction of our review, similar review papers [20], [21], [22], [23], [24] on RUSS are listed in Supplementary Material A, Table S1, including five review papers from different perspectives. Most reviews only focus on the application of the technology in RUSS. For readers interested in autonomous functionalities of RUSS, we recommend References [21] and [23]; for those focused on AI-integrated RUSS, References [20] and [24] provide more detailed insights. To the best of our knowledge, our review is the first to focus on the field of RUSS from a clinical task perspective. The review is organized as follows: Section I provides a brief introduction to US and RUSS. Section II reviews the state-of-the-art advancements in RUSS, providing an overview of current research efforts. It also quantifies the gap between the capabilities of research-driven RUSS and the unmet demands in clinical practice. Section III discusses the organ-specific challenges encountered in RUSS applications and reviews the technological solutions developed to address these issues. Section IV introduces the common challenges faced by RUSS in clinical practice and identifies potential technologies that may help overcome these barriers. Finally, we conclude by outlining the future trends of existing RUSS technologies. These insights are intended to facilitate the integration of RUSS into real-world clinical practice and accelerate their commercialization.

## II. Overview of Robotic Ultrasound Systems

### A. Searching Policy

Driven by the introduction of deep learning around 2006 [25], the field of RUSS has shown steady annual growth in research since 2015. To provide an objective overview of the development of RUSS from 2015 to 2025, we conducted an extensive search of RUSS literature on the Web of Science and Google Scholar. The search terms used were: ('robot∗ AND (ultrasound OR US) AND (learning OR autonomy OR reinforcement OR training)') AND ('imaging OR diagnostics').

We carefully reviewed article titles and abstracts and screened the literature based on the following inclusion and exclusion criteria. Details of the inclusion and exclusion criteria are provided in Supplementary Material A. To ensure a balanced representation of both emerging and mature technologies, we prioritized works with high technical novelty, validated experimental results, and/or demonstrated clinical relevance.

This paper presents inclusive answers to the following Research Questions (RQ) regarding the aims of this research:

*RQ1:* What are the current limitations and research gaps in the development of RUSS for clinical applications?
*RQ2:* Which clinical challenges related to different organs have been addressed by RUSS?
*RQ3:* Why has RUSS still not been used on a large scale in the clinic to date? What are the clinical and technical challenges?

### B. State-of-the-Art

Currently, there is a limited number of RUSS that exhibit strong generalization capabilities and high autonomy across multiple organs. This specialization often arises due to the heterogeneous anatomical and imaging characteristics among different organs and patients, which requires tailored control strategies, imaging techniques, and learning algorithms. For instance, vascular US requires precise Doppler-based imaging for blood flow assessment, while spinal US focuses on structural visualization. The diversity in imaging requirements and clinical objectives poses significant challenges for developing generalized systems. As a result, most existing RUSS are designed and optimized for a single specific body region or organ. Therefore, the RUSS targeting different human organs is classified according to the following criteria:

1) Experiments conducted using human organ phantoms are categorized under the corresponding organ.

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2) Studies involving experiments that directly test on multiple human organs are counted in each relevant organ category.
3) Studies that used empty phantoms or phantoms without specifying a particular organ are also excluded.
4) Organs with a limited number of RUSS studies, such as knee, muscle, and brain, are excluded.
5) RUSS is categorized according to its level of autonomy, with those exhibiting a high level of autonomy being the primary focus of this study.

By comparing the clinical applications of US with those of RUSS, a notable discrepancy emerges: the research priorities of the two fields are not fully aligned. While abdominal US accounts for nearly 69% of all clinical US examinations—according to data collected over the past 15 years in Switzerland [26] —the most extensively studied application domain for RUSS remains vascular imaging, with 54 published studies to date. This misalignment underscores a significant gap between the clinical demand for US imaging and the current research focus on RUSS development. Other clinically significant areas, such as musculoskeletal conditions and endosonography, receive comparatively little attention in RUSS research. This imbalance underscores a critical gap in aligning research priorities with actual clinical needs, suggesting an opportunity for future RUSS development to better address areas of high clinical utility. Nevertheless, it is important to acknowledge that due to the limitations of publicly available data, the conclusions in this section may primarily apply to developed regions, including Switzerland.

Over the past few decades, some RUSS have been developed, with researchers exploring different technical approaches to RUSS. The level of autonomy (LoA) in medical robotics is classified into six tiers based on the degree of automation [27]: No autonomy, Robot assistance, task autonomy, conditional autonomy, high autonomy, and full autonomy. Among these, teleoperated RUSS falls under LoA 1, where the system primarily aids the operator. Most autonomous RUSS are categorized as LoA 2 or LoA 3. Post the COVID-19 pandemic, an exponential surge in the utilization of telerobotic systems has been observed [28]. However, compared to teleoperated RUSS, Autonomous RUSS has the potential to mitigate the instability associated with human operation and reduce variability in decision-making during imaging, offering a promising direction for the medical field. This approach may help address longstanding challenges in US imaging, such as limited reproducibility, the lack of standardized procedures, and the high workload faced by sonographers. In this paper, our focus is on autonomous RUSS, referred to here as RUSS.

Due to the lack of autonomy in teleoperated RUSS, which fundamentally relies on physician operation, its differentiation across various organs is less pronounced. In contrast, RUSS with a high LoA exhibits more significant variations across different organs. By analyzing the discrepancies illustrated in Fig. 2, we identify the following potential reasons:

1) Superficial organs, such as blood vessels, have simpler anatomical structures and are less affected by gas or other internal organs. Experiments targeting superficial organs are safer, pose fewer ethical risks, and are more amenable to technical demonstrations. In comparison, RUSS, designed for obstetrics and gynecology, faces higher operational risks, resulting in slower research progress.
2) Insufficient communication between engineers and clinicians has led to a focus on low-level research (such as control methods based on multimodal information), while studies addressing high-level clinical tasks (such as complete diagnostic and therapeutic workflows) remain relatively scarce.

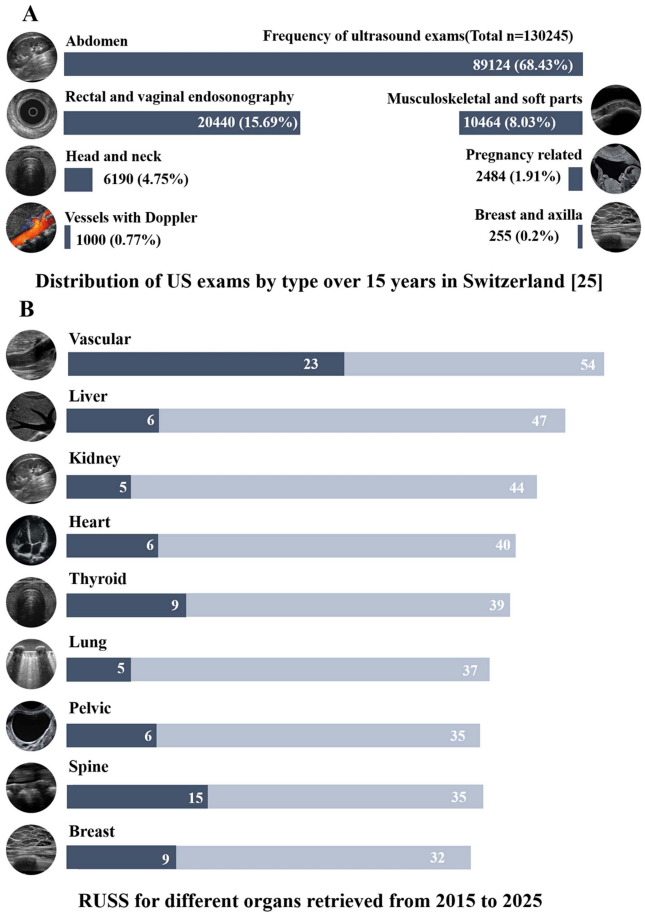



Fig. 2. Distribution of US exams and RUSS research across different organs. While abdominal US dominates clinical applications, comprising nearly 69% of all exams, the majority of RUSS development efforts are concentrated on vascular. (a) Percentage of US examination based on a cross-sectional survey study in Switzerland from 2004 to 2018 [25]. (b) Each bar represents the total number of research papers in RUSS for a specific organ, with the darker section indicating the RUSS that has high level of autonomy. The data were obtained through the retrieval process based on the searching policy outlined earlier.

## III. Organ-Specific Applications of Russ

This subsection introduces RUSS categorized based on their primary objectives, which include the following organ-specific applications: thyroid, vascular, spine, breasts, lungs, heart, abdominal (liver and kidney), and pelvic regions. As illustrated in Fig. 3, RUSS addresses five fundamental tasks designed to overcome key challenges in US imaging.

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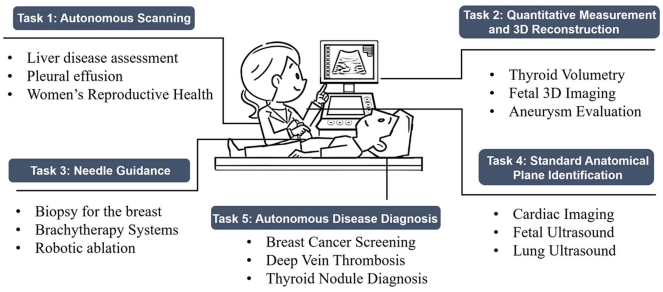



Fig. 3. Fundamental tasks in US imaging.

Task 1: Autonomous Scanning. RUSS represents a transformative advancement in US imaging by enabling reproducible organ coverage, addressing the variability inherent in manual examinations [29]. Through programmable control of scanning trajectories and probe positioning, these systems minimize human-dependent inconsistencies, offering a standardized approach to medical imaging.

Task 2: Quantitative Measurement and 3D Reconstruction. Building on autonomous scanning, RUSS facilitates the automated detection and analysis of structural parameters, converting 2D imaging data into precise 3D morphological representations. For example, vascular examinations benefit from automated measurements of vessel diameter, wall thickness, and plaque areas [30], while thyroid imaging achieves accurate nodule characterization [31]. For readers interested in the conversion of acquired 2D US images to 3D representations, comprehensive reviews are available in [32] for further details on this capability. Recent advances in neural rendering techniques [33], have enabled high-fidelity 3D US reconstruction from sparse 2D acquisitions by learning implicit volumetric representations. This synergy improves 3D reconstruction and offers clinicians reliable, reproducible data to support better diagnostic and treatment decisions.

Task 3: Robotic-assisted Needle Guidance. Needle guidance represents perhaps the most transformative application of RUSS, particularly in minimally invasive interventions [34]. However, traditional manual needle guidance relies heavily on the surgeon's experience and visual perception, making it challenging to accurately position the needle tip and visualize its location [35]. Compared to traditional open surgery, RUSS-guided needle placement offers reduced trauma, faster recovery, and enhanced precision [36].

Task 4: Standard Anatomical Plane Identification. The acquisition of standard US planes plays a crucial role in medical diagnostics [37]. By obtaining standard planes, physicians can reduce measurement errors, enabling accurate assessments of the condition of organs or tissues while also improving the reproducibility of measurements. RUSS leverages advanced computer vision algorithms to perform automatic segmentation and measurement of anatomical structures with high precision [38].

Task 5: Autonomous Disease Diagnosis. The application of RUSS technology in autonomous disease diagnosis represents a significant advancement in medical imaging. By integrating advanced perception capabilities with computer vision algorithms, these systems have the potential to identify subtle pathological changes that may be challenging for human observers to detect [39]. Furthermore, RUSS platforms can generate highly reproducible and standardized imaging results, enhancing diagnostic accuracy and reliability across various medical specialties. The capacity to consistently capture, analyze, and interpret medical images marks a meaningful development in diagnostic methodologies, contributing to more precise and effective clinical decision-making.

Through a comparison of the clinical applications of US and RUSS technologies, we identified a second significant research gap. While clinical US has demonstrated broad utility across a wide range of scenarios, RUSS has yet to reach a comparable level of versatility and effectiveness, particularly in areas beyond Task 1. A detailed illustration of this gap is provided in Supplementary Fig. s1, as it was moved to the supplementary materials due to space constraints. Fig. s1A shows the clinical uses of US as reported by the survey group. Based on a quantitative study [26] analyzing 15 years of billing data from 213 general practitioners (GPs) in Central Switzerland and cross-sectional survey data completed by 61 GPs. Fig. s1B illustrates the specific range and proportion of RUSS applications in different organs. It categorizes the tasks performed by RUSS into five main types. Tasks in the Kidney, Lung, Breast, Heart, and Pelvic regions are limited to T1 and will not be shown additionally.

Table S2 summarizes RUSS applied to organs-related tasks, categorized by task type, specific methods used, and experimental subjects. Learning-based methods, including supervised learning [40], reinforcement learning (RL) [41], and imitation learning (IL) [42], have shown promise in enhancing generalization ability and autonomy. Those tested solely on pilot phantom studies or validated with human volunteers demonstrate varying levels of technological maturity and generalization capabilities of the robots.

### A. RUSS for Thyroid

US is the primary imaging modality for evaluating the thyroid, offering valuable diagnostic insights into thyroid pathologies and adjacent structures [43]. It enables real-time visualization of structures as small as 2 mm, supporting the detection of minute thyroid and parathyroid tumors, as well as assessing thyroid blood flow [44]. However, US findings do not always correlate perfectly with histopathological results, and significant variability exists among experienced sonographers in interpreting thyroid images [45]. Such interobserver differences are critical, particularly when monitoring changes in thyroid nodules over time. These challenges highlight the need for more standardized, reproducible, and automated solutions, which RUSS aims to address.

***1) Task 1 in Thyroid:*** In recent years, RUSS has demonstrated significant progress in enabling autonomous scanning and enhancing operational consistency during thyroid US procedures. Nevertheless, as depicted in Fig. 4(a), the current application scope of RUSS in thyroid imaging remains largely confined to achieving adequate image quality and recognizing

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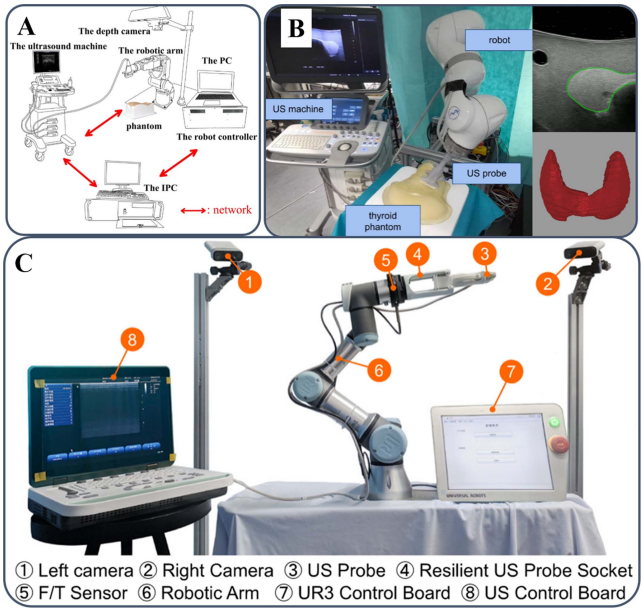



Fig. 4. Representative RUSS in thyroid. (a) RUSS for task 1. Automatic RUSS for 3D imaging [49]. (b) RUSS for task 2. Robotic sonography for thyroid volumetry, online 2D segmentation, and final 3D thyroid segmentation [51]. (c) RUSS for task 5. A fully autonomous RUSS enables an estimate of TI-RADS level with key characteristics of nodules [52].

key anatomical features. Kaminski et al. [46] proposed a method that integrates a fixed scanning path with real-time force feedback control to enhance the consistency of image acquisition. Yet, their validation was restricted to thyroid phantoms, without extending to actual diagnostic evaluations. Similarly, Tsumura et al. [47] contributed a novel robotic end-effector specifically engineered to exert a light contact force (1–2 N), which was tested on healthy volunteers during thyroid scans. Moving beyond hardware-oriented efforts, Zhou et al. [48] employed multimodal information and prior anatomical knowledge to construct an automated system aimed at thyroid nodule detection.

***2) Task 2 in Thyroid:*** Quantitative assessment of thyroid parameters—such as gland length and volume—has emerged as a key area in the advancement of RUSS technologies. As illustrated in Fig. 4(b), Kojcev et al. [12] compared robotic and manual measurements in 144 clinical cases, reporting markedly improved repeatability with the robotic system. Huang et al. [49], [50] developed an automated 3D US imaging framework, achieving a measurement error below 1%. In another notable study, Zielke et al. [51] integrated neural network–based segmentation with autonomous RUSS scanning to perform thyroid volumetry, achieving a mean error of 8.23 $\pm$ 3.10% relative to CT-based ground truth. Collectively, these studies highlight RUSS's potential to improve measurement accuracy and reproducibility, supporting advanced applications like individualized radioiodine therapy.

***3) Task 5 in Thyroid:*** In clinical diagnostics, Su et al. [52] developed a fully autonomous RUSS capable of identifying and grading thyroid nodules. As depicted in Fig. 4(c), the system generates diagnostic outputs consistent with the criteria outlined by the American College of Radiology Thyroid Imaging Reporting and Data System (ACR TI-RADS). While these initial results are promising, further large-scale clinical validation remains necessary to assess the system's robustness and generalizability across diverse patient populations.

Despite the above advancements, current RUSS for thyroid examination is limited in their ability to meet clinical demands. Most systems remain constrained to imaging and measurement tasks, with limited validation in real-world clinical settings. Future research should focus on integrating advanced diagnostic algorithms, expanding autonomous capabilities, and conducting large-scale clinical studies to bridge the gap between technological innovation and clinical utility.

## *B. RUSS for Vascular*

Diagnostic assessment, staging, and imaging of suspected arterial diseases form the cornerstone of successful treatment. Non-invasive vascular assessments serve as an extension of the clinical history and physical examination for vascular diseases, aiding in the confirmation of arterial disorders and the evaluation of lesion location, extent, and severity [53]. Current assessment methods include physiological tests and duplex ultrasonography (DUS). The former determines the location and severity of arterial occlusive lesions based on symptoms, while the latter provides a more precise evaluation of vascular anatomy. DUS not only assesses the anatomical location and extent of vascular lesions but also evaluates flow velocity and volume [54]. Most RUSS are developed for blood vessels due to their simple anatomy. An overview of RUSS applications for Vascular is presented in Table S2.

***1) Task 1 in Vascular:*** US-based vascular imaging and measurement are critical in both diagnostic evaluation and robotics-assisted vascular interventions [55]. The autonomous execution of staged vascular scanning has been validated across multiple studies, as illustrated in Fig. 5(a) [56]. For instance, Huang et al. [57] proposed a procedure-specific learning framework informed by standard clinical protocols to facilitate autonomous carotid artery scanning. Their approach decomposes the aortic assessment into three sequential stages and four operational states, enabling RUSS to capture US views with distinct and clinically meaningful characteristics. In a related advancement, Chen et al. [58] developed an automated system for 3D femoral artery US acquisition, enabling precise vascular localization and 3D reconstruction for preoperative planning.

Given that sudden patient movements or doctor-directed pose adjustments can lead to imaging errors or interruptions, recent advancements in motion-aware RUSS aim to address clinical challenges such as patient movement and anatomical variability, both of which degrade 3D US compounding and trajectory planning. Yan et al. [59] introduced a multi-modal interaction control scheme for RUSS designed to collaborate closely with a doctor in a shared environment. Experimental results show that the system maintains both safety and functionality, even in challenging scenarios such as patient movement, doctor involvement, or other unexpected events. Three studies propose vision-based solutions to enable real-time motion monitoring and adaptive trajectory adjustment. Jiang et al. [60] employed depth cameras

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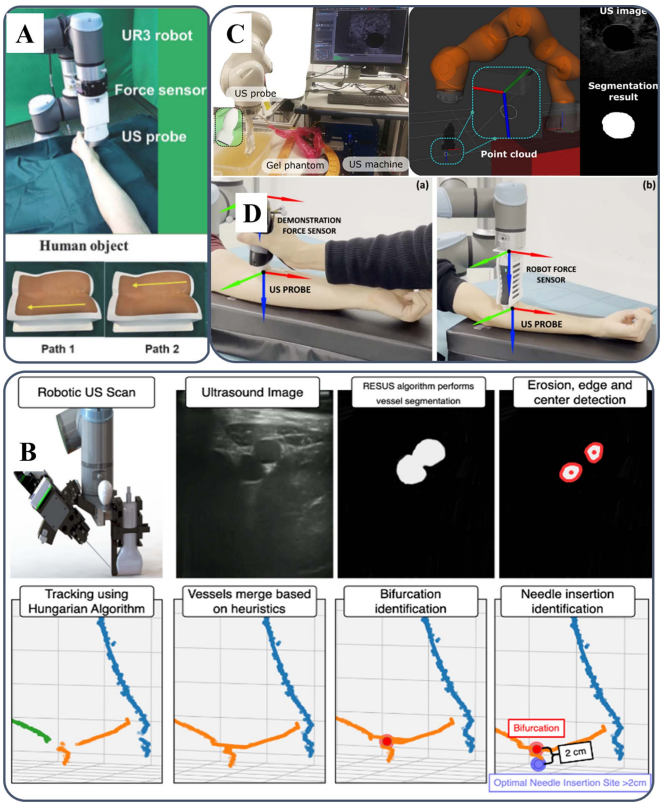



Fig. 5. Representative RUSS in vascular. (a) RUSS for task 1. Autonomic RUSS scanning in an unstructured environment [56]. (b) RUSS for task 3. Bifurcation Identification For US-driven robot cannulation [73]. (c) RUSS for Task 2. RUSS enables accurate calculation of vessel radii [68]. (d) RUSS for Task 5. RUSS for DVT US examination [81].

to track predefined trajectories and passive markers for motion compensation, while they [61] also integrated RGB-based object segmentation with iterative closest point (ICP) registration to recompute trajectories post-movement. In addition, They [62] developed MRI-derived vascular atlases to project preplanned paths onto patient-specific surfaces, enhanced by a U-Net network to improve spatial continuity in vessel reconstruction. Tissue deformation in US imaging leads to geometrical errors when measuring tissues due to the pressure exerted by US probes. Jiang et al. [63] developed a deformation-aware RUSS to correct the tissue deformations in robotic 3D US acquisitions. DefCor-Net [64], proposed by Jiang et al., enabled the potential for anatomy-aware deformation correction.

***2) Task 2 in Vascular:*** In medical image analysis, 3D quantitative measurements—such as length, angle, area, and volume—are essential for accurately characterizing of anatomical structures and the investigation of organ-level physiological dynamics [65]. These measurements serve as critical inputs for evidence-based clinical decision-making [66]. To quantify vascular radius, Ma et al. [67] introduced an autonomous US-guided robotic system designed to measure the radius of peripheral vessels, providing a detailed assessment of vascular structures. As shown in Fig. 5(c), Jiang et al. [68] proposed an automated RUSS workflow for screening tubular structures based on real-time US feedback. Their optimization strategy allows for precise estimation of vessel radii, even under probe orientations deviating from the surface normal. Additionally, Huang et al. [69] introduced a dedicated RUSS pipeline for carotid imaging, aimed at improving the accuracy and reproducibility of vascular assessments. Collectively, these efforts highlight the growing potential of RUSS in enhancing the fidelity and efficiency of vascular US imaging.

Notably, the volume and cross-sectional areas of plaques are critical for evaluating disease progression and informing clinical management [65], as exemplified by measurements of carotid plaques. Despite numerous studies utilizing artificial intelligence for post-processing of 3D carotid US to quantify plaques [70], research employing RUSS to complete the full workflow from scanning to measurement remains limited. The targeted development of RUSS to address clinical needs is crucial, as it directly determines its clinical relevance and utility.

***3) Task 3 in Vascular:*** Several studies have addressed task-specific challenges in RUSS, including precise needle insertion, differentiation of arteries and veins, and the enhancement of procedural safety. Accurate vascular identification in US imaging is particularly critical for guiding vascular interventions in robot-assisted procedures. To this end, Geng et al. [71] proposed a segmentation strategy that incorporates force-sensing feedback into the RUSS framework, improving artery-vein discrimination by exploiting their differing mechanical properties. Koskinopoulou et al. [72] advanced this field by developing an autonomous dual-robot system for venous access, which combines real-time US and bioimpedance sensing. Through precise co-registration, this system facilitates accurate needle insertion, enables robust visualization of deep vessels, and ensures reliable venipuncture detection. Vessel bifurcations serve as key anatomical landmarks, are instrumental for guiding safe catheter or needle placement. Morales et al. [73] introduced a bifurcation-aware algorithm for US-guided robotic cannulation, which identifies optimal insertion sites by integrating domain expertise with deep learning techniques. Validated in live pig models, this system demonstrated precision comparable to experienced clinicians, as illustrated in Fig. 5(b).

***4) Task 5 in Vascular:*** A subset of studies has focused on developing RUSS for the diagnosis and monitoring of vascular diseases. In the context of peripheral arterial disease (PAD), accurate localization and severity of arterial stenosis are essential for guiding appropriate clinical interventions [74]. Merouche et al. [75] introduced a RUSS using B-mode imaging to continuously track and reconstruction of the lower limb arterial tree from the groin to the knee. For abdominal aortic aneurysm (AAA), a pathological dilation of the aorta with high rupture risk, clinical management requires precise quantification of vessel morphology and biomechanical stress [76]. To support AAA screening, Virga et al. [77] developed an autonomous RUSS designed to perform quantitative vessel diameter measurements. Addressing challenges related to respiration-induced motion artifacts in 3D US reconstruction, Velikova et al. [78] proposed a compensation strategy based on implicit neural representations, enhancing spatial accuracy.

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Deep venous thrombosis (DVT), another prevalent vascular pathology, involves the formation of blood clots in deep veins and may lead to pulmonary embolism (PE), a potentially fatal complication [79]. Despite the widespread use of US in DVT diagnosis, diagnostic performance remains heavily dependent on operator skill. Due to frequent misdiagnosis, the actual prevalence of DVT and PE is difficult to determine, with studies suggesting that up to 60% of untreated DVT cases may progress to PE, contributing to over 50000 deaths annually [80]. To reduce this variability, Dall'Alba et al. [81] proposed a learning-based approach that trains autonomous robotic controllers on sonographer demonstrations, thereby standardizing DVT US assessments. This approach is illustrated in Fig. 5(d). Complementarily, Huang et al. [82] developed a semi-autonomous RUSS equipped with an advanced physical human-robot interaction (pHRI) interface based on virtual fixture techniques, enabling clinicians to quantify thrombus extent with enhanced precision and consistency.

Femoral artery examinations require long-range scanning from the inguinal region to the proximal thigh. Manual operation often leads to image discontinuity, unstable probe pressure, and positioning errors, compromising consistency. RUSS offers stable execution of large-scale linear or curved scanning trajectories, ensuring continuous and high-quality imaging [83]. Carotid artery examinations emphasize the repeatability of plaque detection, intima-media thickness measurement, and hemodynamic assessment, which are highly susceptible to operator variability. RUSS, with standardized scanning paths and precise force control, significantly improves image acquisition consistency, making them well-suited for long-term follow-up and multicenter studies [84].

Although existing systems have addressed autonomous scanning and imaging of carotid and femoral arteries, they often focus on isolated aspects rather than providing an integrated solution for complete vascular assessment. However, there are few clinical applications for vascular imaging alone, which limits the application of RUSS. Current RUSS implementations generally lack the adaptability and comprehensive functionality required for routine clinical adoption. This disconnect suggests a critical area for future research, particularly in developing more versatile, integrated systems that can handle the full spectrum of vascular imaging requirements while maintaining robustness in real-world clinical settings.

### C. RUSS for Spine

US is currently in the stage of prospective research in spine surgery, being explored for its potential in measuring the spinal canal, detecting cord abnormalities, diagnosing scoliosis [85] and examining soft tissue [86], as illustrated in Fig. 6(a). Its advantages include non-invasiveness, portability, and real-time imaging. Recent advancements in US technology have expanded its applications, such as quantifying scoliotic curves [87], measuring multifidus muscle size [88], and imaging sciatic nerve lesions [89]. Clinically, US offers significant benefits in both diagnosis and treatment: it provides clear visualization of muscles, tendons, and joints for early detection of spinal diseases, while

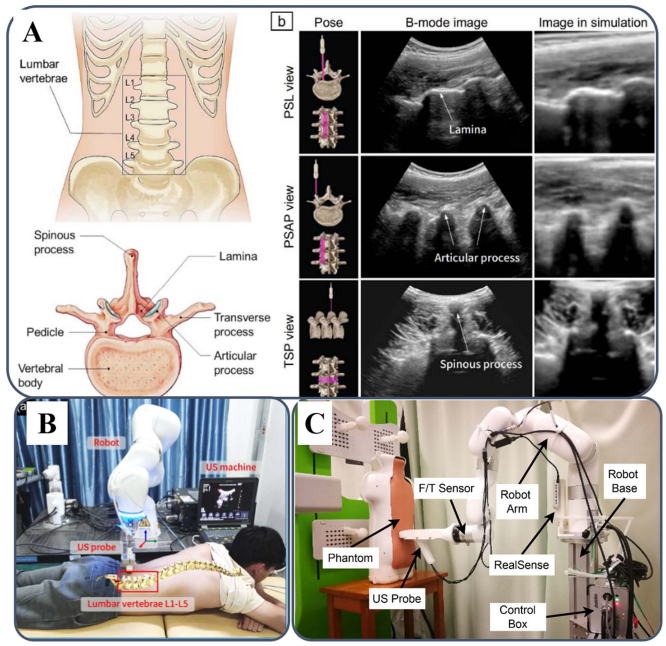



Fig. 6. Representative RUSS in spine. (a) Lumbar spine anatomy and US acquisitions of three standard views of the lumbar spine [95]. (b) RUSS for task 2. RUSS is used to acquire US images of the L1-L5 lumbar vertebrae of the volunteers for 3D volume reconstruction [93]. (c) RUSS for Task 5. RUSS follows the spinal curvature by real-time detection of spinous processes on US frames [96].

US-guided interventional therapies, such as selective nerve root block and facet joint block, have shown high accuracy, safety, and efficacy [90]. However, challenges such as artifacts from spinal bones and the complexity of spine anatomy limit its broader adoption in clinical practice.

***1) Task 1 in Spine:*** To improve image quality, Jiang et al. [91] proposed an approach for optimizing probe orientation by aligning the central axis of the probe with the surface normal of the spinal tissue. In a subsequent study expanding upon this method, the same group [92] achieved automation of this alignment process, thereby enabling accurate and consistent control of the probe orientation relative to the spinal surface. Further advancing this line of research, Li et al. [93] introduced a control framework for autonomous adjustment of a virtual US probe's six degrees of freedom (DOF) pose, utilizing real-time image feedback to guide the probe toward standard scanning planes. As illustrated in Fig. 6(b), a RUSS system is shown acquiring B-mode images of the L1-L5 lumbar vertebrae in human volunteers, which are subsequently reconstructed into 3D volumes. To address the mechanical complexities inherent in spinal scanning, Wang et al. [94] developed a RUSS with compliant joints tailored for spinal imaging. In addition, Hase et al. [97] extended the scope of US-guided robotic navigation to include sacral imaging, thereby further demonstrating the adaptability and potential of RUSS technology in spine-related clinical applications.

***2) Task 3 and Task 4 in Spine:*** Despite recent advancements in imaging technologies, significant challenges remain in clinical translation of technical innovations. For instance,

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although US provides a radiation-free alternative to X-ray imaging for guiding lumbar spinal injections, its clinical utility remains constrained by bone-induced artifacts and soft tissue noise [98]. These challenges have led to advanced image processing techniques to improve vertebral level detection. In response to this need, Tirindelli et al. [99] proposed a RUSS-based method for automatic detection of vertebral levels. This system autonomously conducts spinal scanning by leveraging fused sensory data—integrating US and force information—to provide both tactile and visual feedback for localization. In parallel, Li et al. demonstrated autonomous acquisition of standard spinal imaging planes. Specifically, they [95] introduced an innovative dual-agent framework that jointly optimizes probe motion using real-time US imagery, simulating expert-level decision-making to achieve consistent acquisition of standard views in spinal sonography.

***3) Task 5 in Spine:*** Adolescent idiopathic scoliosis (AIS) is a common spinal condition, with an estimated incidence of 0.47% to 5.2% in adolescents [100]. The standard Cobb angle measurement requires repeated X-rays, resulting in cumulative radiation exposure. Although low-dose EOS imaging has been introduced to alleviate radiation-related concerns, its widespread adoption is hindered by substantial cost, particularly in settings with limited medical resources [101]. In contrast, US offers a safe, affordable, and radiation-free option for routine scoliosis monitoring [102]. The feasibility of utilizing three-dimensional US for scoliosis evaluation was demonstrated by Zheng et al. [103]. However, to date, no 3D US system with validated reliability and clinical readiness for scoliosis assessment has been reported. Victorova et al. [104] proposed a RUSS methodology for tracking spinal curvature and enabling automated navigation. As depicted in Fig. 6(c), Duan et al. [96] explored the design of an optimization-driven control strategy incorporating variable impedance to facilitate US-guided scoliosis assessment via an assistive robotic system. Furthermore, three studies [50], [107], [108] have addressed technical innovations such as RL-based control and robotic mechanical design. Nevertheless, these efforts have yet to fully account for the unique anatomical characteristics of the spine, particularly in cases involving structural deformities.

## D. RUSS for Abdominal

US is, in most cases, the first imaging technique used to diagnose and characterize abdominal pathology [109]. The abdomen is the part of the body between the chest and the pelvis. Organs in the abdomen include the liver, gallbladder, pancreas, spleen, and kidneys. The abdominal US shows the size of organs, morphology, blood flow, and if there is injury or disease.

Autonomous US scanning (Task 1) has been most extensively studied in abdominal applications, particularly for liver and kidney examinations.

For instance, in hepatology, US is employed to assess liver size, echotexture, and the presence of lesions such as tumors or cysts. In nephrology, US evaluates kidney morphology, detects renal stones, and measures renal artery blood flow to diagnose vascular abnormalities. This section will focus on RUSS designed specifically for liver and kidney scanning.

***1) RUSS for Liver:*** For the liver, US is crucial in detecting a wide range of conditions, including fatty liver disease, fibrosis, hepatic tumors, and vascular abnormalities [110]. It enables precise characterization of lesion morphology, differentiating between benign and malignant masses, and assessing liver parenchyma texture. Due to the acoustic shadow cast by the subcutaneous rib cage [111], the US examination is still challenging for internal organs, such as the liver and heart, which are covered by rib cages.

Most existing RUSS for liver examination concentrate exclusively on autonomous scanning protocols. Bi et al. [112] proposed a method to tackle the challenging task of intercostal US scanning path planning. Zhang et al. [113] proposed a novel autonomous RUSS targeting liver follow-up scans for outpatients in local communities. Mustafa et al. [114] proposed a RUSS for automatic US imaging focusing on the human liver, which is composed of four elements. Ipsen et al. [105] proposed a RUSS with dynamic force control for stable, long-term imaging of anatomical regions affected by motion. The system facilitates the acquisition of 4D image data in vivo over extended scanning periods for the first time.

US plays a crucial role in image-guided therapies such as liver ablation. However, the significant acoustic impedance mismatch between bone and soft tissues necessitates accurate probe positioning between the ribs to ensure optimal acoustic access. During ablation, precise mid-intercostal insertion of electrodes is essential to prevent vessel injury [115]. The proximity of hepatic tumors to major blood vessels or heat-sensitive organs further underscores the need for highly accurate intervention trajectory planning. Patient-specific anatomical variations pose major challenges to individualized trajectory planning, particularly in thoracic cases with limited acoustic windows, such as intercostal liver imaging [116]. To accurately map planned paths from a generic atlas to individual patients, Jiang et al. [116] presented a class-aware cartilage bone segmentation network with geometry-constraint post-processing to capture patient-specific rib skeletons. Then, they [117], [118] proposed a dense graph-based non-rigid registration to transfer planned paths from the atlas to the current setup by explicitly considering subcutaneous bone surface. Mohan et al. [119] demonstrated the application of robotic US guidance for tumor biopsy procedures. They presented a fully autonomous system for US-guided needle placement capable of performing end-to-end clinical workflow, which has the potential for accurately targeting tumors larger than 5 mm. Despite advances in autonomous intervention, current systems lack integrated diagnostic capabilities, limiting their clinical applicability.

Tumors located at the inferior margin or lower right lobe of the liver can be more readily accessed and imaged via the sub-costal approach, which offers greater maneuverability and reduced bony obstruction compared to the intercostal route. Despite its clinical potential, the sub-costal approach has not yet been explored in the design of RUSS for liver imaging. In other clinical applications, such as robot-assisted lobectomy, the sub-costal channel has been successfully employed for specimen retrieval, demonstrating the feasibility of this route in robotic procedures [120]. These findings suggest that the sub-costal

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approach holds potential for future applications in RUSS-based liver imaging.

***2) RUSS for Kidney:*** A variety of imaging modalities are available for the evaluation of kidney diseases. Among these, renal US is the most commonly used in patients with kidney conditions due to its safety, ease of use, and reliable results. Kidney US provides essential insights into renal health by screening for cysts, stones, and tumors while also assessing kidney size, cortical thickness, and potential obstructive uropathies. This technique is particularly effective in detecting hydronephrosis, identifying renal calculi, and monitoring the progression of chronic kidney disease. Its real-time imaging capability allows dynamic assessment of organ function, blood flow, and structural integrity.

Similarly, existing RUSS for kidney examination concentrates exclusively on autonomous scanning protocols. Deng et al. [106] proposed to learn the US scanning skill in kidney phantom from human demonstrations, which takes the US images, pose/position of the probe, and contact force into account. Wang et al. [121] proposed a general control framework for task space-compliant motion and 6D force regulation towards automated US imaging. Two studies [56], [108] have primarily emphasized technical implementations, and while they utilized kidney phantoms for testing, their approaches were not specifically designed to address the unique anatomical structures of the kidney.

Deep abdominal organs like the liver and kidneys present unique challenges due to their anatomical complexity and individualized environment. For the liver, the acoustic shadow caused by the rib cage complicates intercostal scanning [122], and current RUSS systems primarily focus on autonomous scanning protocols without incorporating diagnostic capabilities or addressing disease-specific imaging requirements. Similarly, kidney-targeted RUSS, while effective for automated scanning, lacks adaptability to the kidney's distinct anatomical features and often fails to account for individualized physiological conditions such as Body Mass Index (BMI) [123] of different patients and patient sudden movement. Furthermore, most systems emphasize technical implementation over integration into a complete clinical workflow, including diagnostic interpretation and decision-making. Addressing these gaps requires the development of adaptive frameworks tailored to deep abdominal organs, incorporating patient-specific models, real-time force and motion regulation, and diagnostic functionalities to enhance the clinical utility of RUSS in abdominal US.

## E. RUSS for Other Organs

Some organs, such as the lungs, breasts, heart, and pelvic region, have seen limited development in RUSS, with most systems exclusively focusing on the task of autonomous scanning. Due to the relatively small number of studies and the narrow scope of tasks addressed, RUSS targeting these organs is grouped into a single category for analysis. It is noteworthy that US imaging targeting these organs, particularly the breast, plays a pivotal role in clinical practice. However, the development of RUSS for these applications remains relatively limited. An overview of RUSS applications for these organs is presented in Table S2.

***1) RUSS for Lung:*** Lung US (LUS) is a useful way for diagnosis and severity evaluation of respiratory diseases [124]. US provides dynamic visualization of lung tissue, allowing clinicians to detect subtle changes in lung architecture, pleural effusions, and interstitial syndrome. LUS is particularly valuable in critical care settings, enabling rapid bedside assessments of respiratory conditions such as pneumonia, lung consolidation, pneumothorax, and pleural fluid accumulation [125], [126]. By utilizing high-frequency transducers, clinicians can observe lung sliding, B-lines, and pleural movement, which provide critical diagnostic insights into respiratory pathologies. LUS's portability, cost-effectiveness, and ability to perform serial examinations make it an increasingly essential tool in both acute and chronic respiratory disease management [127]. Despite these benefits, it is still challenging to obtain consistent image quality and maintain precise placement of the probe, especially in dynamic environments where human breathing is constantly changing.

Recent advancements in RUSS aim to address these challenges by enhancing imaging automation in LUS. Ma et al. [128] introduced a RUSS framework that employs an RGB-D camera and DensePose to identify scanning targets on the patient. The system automatically attaches the probe vertically to the tissue surface based on the calculated landing posture. However, the workflow lacks detailed descriptions of global trajectory planning and its adaptability to diverse patient anatomies. To improve probe alignment during imaging, they [129] also designed an end-effector equipped with laser sensors, enabling real-time adjustment to maintain correct probe positioning. Integrated with the RUSS system, this approach demonstrated LUS image quality comparable to manual scanning. Building on this foundation, they [130] proposed a standardized imaging plane (SIP) navigation framework for robotic LUS. By extracting multiple anatomical features from real-time LUS images and utilizing non-patient-specific template matching, the system achieves automated probe adjustments toward the SIP via image-based visual servoing (IBVS). This framework facilitates accurate probe navigation but may face challenges in adapting to anatomical variability across patients. In addition to probe alignment, advancements have been made in operator-assistance frameworks. Nardi et al. [131] presented an anatomy-aware control framework for teleoperated LUS. Using biomechanically accurate 3D models, the system provides real-time visual feedback and applies virtual constraints to assist in precise probe placement tasks. Furthermore, Tan et al. [132] introduced a fully automatic dual-probe scanning robot optimized with a velocity look-ahead strategy to enhance trajectory planning. Their approach achieved stable contact force and improved scanning efficiency, but its complexity may limit its application in routine clinical practice.

Gaps remain in meeting the clinical requirements specific to LUS. Current systems primarily focus on automation and image quality but struggle to adapt to the unique challenges posed by lung anatomy and its dynamic nature. For instance, the constant

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movement of the lungs due to respiration introduces additional variability, which existing frameworks, like SIP navigation, cannot fully accommodate as they rely on non-patient-specific templates. Furthermore, real-time adjustments remain inadequate, systems such as IBVS, while effective in maintaining probe alignment, lack the responsiveness needed to adapt to respiratory motion, patient movement, or sudden changes in lung conditions.

***2) RUSS for Breast:*** The latest global cancer burden data released by the International Agency for Research on Cancer in 2020 show that breast cancer has replaced lung cancer as the highest-incidence cancer worldwide [133]. Breast US represents a critical non-invasive imaging modality that complements mammography in detecting and characterizing breast pathologies. Unlike radiation-based imaging techniques, US offers real-time, radiation-free visualization of breast tissue, making it particularly valuable for younger patients and those with dense breast composition [134]. RUSS can precisely map breast tissue, detect subtle lesions, and characterize tumor morphology with unprecedented accuracy. By automatically tracking lesion boundaries, measuring size and vascularity, and generating standardized imaging reports, RUSS technologies transform breast imaging from a subjective to an objective diagnostic process. The ability to perform consistent, reproducible scans across different examination sessions provides clinicians with reliable comparative data, enhancing early detection capabilities for conditions ranging from benign cysts to malignant tumors.

However, a majority of RUSS developed for breast imaging focus on achieving autonomous scanning functionality. Welleweerd et al. [135] presented how an autonomous 3D breast US acquisition can be performed utilizing a 7-DOF robot equipped with a linear US transducer. Expanding on previous work, they [136] investigated how to change the direction of the scanning trajectory based on US feedback, such that no patient-specific model is required to perform a scan. Tan et al. [137] proposed and implemented a probe automatic trajectory planning system for US imaging based on 3D point cloud, which realizes the global automatic generation of probe scanning trajectory. Furthermore, They [133] proposed a flexible and autonomous US scanning system for automatic breast US imaging. The system integrates the task of 3D perception, the task of trajectory planning, and the task of closed-loop force control end-to-end, but the details of trajectory generation are not introduced. Wang et al. [138] aimed to solve the problems of path planning and interaction control in robotic US scanning of breasts.

A notable advantage of RUSS is its ability to transform breast imaging into a standardized process, providing consistent and reproducible scans across sessions. This potential could enhance longitudinal monitoring of breast conditions, particularly in high-risk populations. However, Breast anatomy and pathology can vary widely [139], and current trajectory planning methods often neglect these variations, potentially limiting diagnostic accuracy. Current RUSS for breast imaging focuses heavily on achieving autonomous scanning functionality. The development of RUSS for breast imaging should move beyond basic scanning functionalities to address broader clinical challenges. Integration of lesion classification algorithms and elastic US can make RUSS a more comprehensive tool.

***3) RUSS for Heart:*** Cardiac US, also known as echocardiography, is a non-invasive imaging technique that evaluates the structure and function of the heart. Echocardiography can visualize the size of the heart chambers, the thickness of the heart walls, the movement of heart valves, as well as the speed and direction of blood flow. It is an essential tool for diagnosing various cardiac conditions, such as valvular heart disease, congenital heart defects, and myocardial diseases [140]. The application of cardiac-oriented US robots is likewise limited to autonomous sweeps, even if only up to tele-ultrasound and semi-autonomous sweeps. Avgousti et al. [141] proposed an end-to-end mobile tele-echography platform using a portable robot for remote cardiac ultrasonography. Kim et al. [142] developed a semi-autonomous RUSS. Their approach combines force feedback, PID control, and CNN-based image classification to guide probe movement, offering a novel solution for automated US imaging. Giuliani et al. [143] presented the collected findings of a user-centered approach for developing a teleoperated robot for remote echocardiography examinations. Jiang et al. [144] presented a Cardiac Copilot system capable of providing real-time probe movement guidance to assist less experienced sonographers in conducting freehand echocar diography. However, it has not been applied to the robotic arm to complete autonomous scanning.

***4) RUSS for Pelvic:*** Pelvic US, with a particular emphasis on bladder imaging in our paper, is a critical non-invasive diagnostic technique that provides comprehensive insights into urological and reproductive system health [145]. The bladder, a dynamic organ that undergoes significant volume changes, requires precise imaging to assess its morphology, wall thickness, and potential pathological conditions [146]. RUSS can systematically evaluate bladder volume, detect wall irregularities, identify potential tumors, and assess post-void residual urine volumes. Beyond basic structural analysis, advanced RUSS technologies can now perform real-time dynamic assessments, tracking bladder filling and emptying processes, which is particularly valuable in diagnosing conditions such as urinary retention, bladder outlet obstruction, and various inflammatory or neoplastic disorders.

The simple anatomy of the bladder as a superficial organ makes it easy to use as an experiment for RUSS sweeps. Santos et al. [147] proposed a new control approach for tele-echography RUSS. The control architecture has been assessed in a clinical scenario, with a gynecologist performing a robotic-assisted pelvic US examination. Raina et al. [148] proposed a sample-efficient Bayesian Optimization (BO) framework using deep kernel and image quality estimators for optimizing the 6D robotic US controller. Building on previous work, They [149] proposed a sample-efficient method to automatically adjust the orientation of the US probe normally to the point of contact on the scanning surface, thereby improving the acoustic coupling of the probe and resulting image quality. Based on the above Research, Raina et al. [150] presented a coaching framework for RUSS to improve the learning efficiency and accuracy of US image acquisition. Unlike previously proposed LfD methods for RUSS, this framework leveraged real-time feedback from experts during the training process.

Prostate biopsy and brachytherapy are two distinct yet complementary procedures in prostate cancer management.

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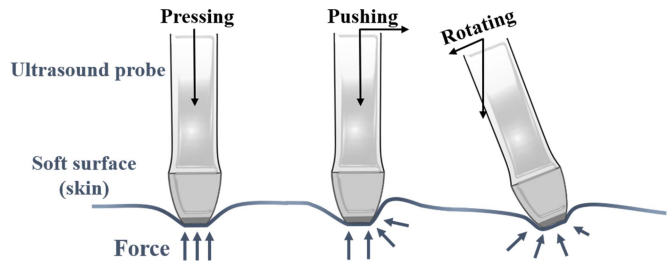



Fig. 7. Surface deformation leads to difficulties in force analysis during US scanning processes.

Biopsy involves inserting a needle to extract tissue samples for diagnostic purposes, whereas brachytherapy entails implanting radioactive seeds directly into the prostate to deliver localized radiation therapy. Both procedures benefit from RUSS, which enhance precision and reproducibility. For instance, Bogdan et al. [151] developed a robotic device for prostate biopsy at autonomy level 1, enabling targeted sampling through US guidance combined with MR image fusion. In another study, Nikolai et al. [152] introduced a RUSS for prostate brachytherapy, notable for its real-time tracking of prostate motion to dynamically adjust dose planning and needle placement during the procedure.

## IV. Overall Challenges of RUSS for Clinical Applications

In the manufacturing industry, robots are generally fixed in specific locations, where they repeatedly execute identical programmed tasks within a controlled environment [153]. However, in dynamic environments that require human-robot interaction, autonomous US scanning poses a non-procedural challenge that cannot be resolved through static algorithms or predetermined steps. It is more complex than a simple 'move from A to B' path-planning problem. This challenge requires robots to handle uncertainties and adapt their actions accordingly [154]. Focusing specifically on RUSS applications, it presents unique challenges in clinical applications, including (a) safety of pHRI, (b) adaptability to the clinical environment, (c) limitations of US data and imaging manipulation, (d) generalization issues, (e) co-design of hardware and software systems, (f) ethical considerations in the deployment of RUSS and (g) regulatory and reimbursement considerations.

### A. Safety of pHRI

RUSS must frequently and continuously interact with the surface of the human skin tissue during task execution. In such contact-rich environments, the contact state is highly dynamic, complex, and constrained [155], [156]. RUSS is required not only to maintain stable contact with the patient's skin tissue but also to precisely control the magnitude and direction of the contact force to achieve uninterrupted US imaging. The intricate pHRI and inherent uncertainties between RUSS and human skin tissue limit the widespread adoption of automated US scanning applications [157]. Tsumura et al. [47] made mechanical innovations by designing a robotic end-effector capable of applying minimal contact force (1–2 N) during thyroid US imaging on human volunteers. IL enables RUSS to mimic the interaction behaviors of sonographers by replicating their force application, thereby capturing aspects of pHRI. Dall'Alba et al. [81] applied IL with Kernelized Movement Primitives (KMP) to train an autonomous robotic controller for standardized DVT US exams, capturing and generalizing sonographer scanning skills, including trajectory and force. In US scanning tasks, the physical interactions often exhibit variability and continuity, making it challenging for traditional fixed-parameter control methods to adapt to these dynamic requirements. Variable impedance control allows robots to respond to interaction forces through compliant behavior. Santos et al. [147] presented a variable stiffness control method for precise directional control and compliant localization that is adequate for this type of teleoperation task.

### B. Adaptability to the Clinical Environment

Human tissue represents a highly unstructured environment characterized by significant uncertainty, variability, and complexity. Its objects, states, and environmental features are challenging to fully describe using deterministic mathematical models [158]. The dynamic nature of the scanning environment arises from continuous tissue motion due to physiological activities (e.g., respiration and heartbeat), involuntary patient movements, and unpredictable tissue deformation and displacement. Additionally, the ambiguity and diversity of tissue interfaces, spatial heterogeneity of acoustic properties, and fluctuations in image quality due to various factors further exacerbate environmental uncertainty. However, most RUSS [159], [160] have been validated only on phantoms, with limited verification of healthy volunteers.

Compared with traditional scanning approaches that require continuous transducer movement, integrating 3D US devices with RUSS to perform fixed-point volume acquisition is technically less demanding and may be more feasible in certain clinical environments. In such systems, the robot is primarily responsible for positioning the 3D probe at an optimal location, while the US device captures a volumetric image without requiring probe movement. This simplifies the robotic task and enables fully automated 3D image acquisition, which can be seamlessly combined with existing AI-based diagnostic tools. Although this approach may not be suitable for all anatomical regions, it offers significant promise for applications involving stationary organs or easily accessible body sites. For example, The Automated Breast Volume Scanner (ACUSON S2000 ABVS; Siemens Medical Solutions, USA) can scan the entire breast volume by generating consecutive sonographic images [161]. The working process includes automatic breast volume scanning, data acquisition, and image processing into 3D images that can be analyzed in any desired direction.

### C. Limitations of US Data and Imaging Manipulation

US images exhibit inherent challenges for both clinical practice and automated systems. The complexity of US image interpretation stems from its characteristic speckle patterns, acoustic

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shadows, and artifacts, which often obscure anatomical structures and pathological features [111], [162]. Furthermore, the restricted field of view in US imaging provides only limited spatial context, necessitating continuous mental reconstruction of the complete anatomical structure by the operator [163]. The combination of complex image content and limited spatial coverage impacts diagnostic accuracy and increases the learning curve for novice practitioners, ultimately affecting the reproducibility and reliability of US examinations.

US scanning, as a typical non-procedural task, encompasses an integrated workflow comprising perception, planning, control, and decision-making. Patient-specific variability presents significant challenges to this type of task. In the perception stage, the instability of low-quality US images affects the accuracy of feature extraction as well as the localization and tracking of anatomical landmarks. Additionally, the fusion of multi-source heterogeneous data, such as US video, force feedback, and kinematic information, further increases the complexity of visual perception. In the planning stage, pre-defined robotic arm trajectories struggle to adapt to dynamic environmental changes [164]. Moreover, probe path planning must account for multiple constraints and safety boundaries. Dynamic position errors caused by continuous tissue deformation also complicate the control of probe contact force and orientation with the skin. In the decision-making stage, there is a need to address multi-objective optimization challenges, such as balancing image quality and operational safety. Furthermore, standardizing, quantifying, and modeling expert knowledge remains a key obstacle.

Additionally, during US scanning, sonographers often explore different perspectives and positions to locate targets. Due to suboptimal search strategies or environmental uncertainties, non-optimal operations may occur, resulting in the "suboptimal demonstration" problem [165], [166]. An agent typically learns a specific task by observing and mimicking expert behaviors in IL. However, if the provided demonstrations are flawed or not best practices, the learned strategies may also be suboptimal, limiting the agent's performance and, in some cases, leading to undesirable outcomes.

### D. Generalization Issues

Although RUSS targeting different anatomical structures generally share similar frameworks and system designs, few systems demonstrate robust generalization performance and have been validated on multiple real patients. While some studies [167] have conducted experiments on multiple phantoms, the gap between phantoms and real human anatomy remains significant. The RUSS, when removed from the clinical environment, has certain limitations. Some defects that may be obscured in phantom experiments are exacerbated in real clinical scenarios. For example, the nonlinear friction characteristics of robotic arm joints have minimal impact in static phantom scans, but they may accumulate into significant trajectory deviations during respiratory motion compensation tasks. Similarly, surface reconstruction algorithms based on RGB-D cameras perform well on phantoms with uniform skin tones but may introduce registration errors when applied to patients with skin pigmentation or hair interference.

Furthermore, anatomical variations among patients with different BMI, age, and gender introduce significant individual differences. Robust generalization is a critical prerequisite for the clinical application of RUSS, requiring consistent performance across a wide variety of experimental subjects. It's worth noting that pathological conditions can induce varying degrees of morphological changes in human tissues, further challenging the adaptability of RUSS in diverse and complex clinical scenarios [168]. Moving forward, RUSS must be extensively validated on both healthy volunteers and patients to ensure their suitability for real-world clinical applications.

### E. Co-Design of Hardware and Software Systems

The clinical translation of RUSS is not only limited by the performance of individual hardware or software components but also faces systemic challenges related to hardware-software co-design. Industrial robots typically employ tightly coupled hardware-software architectures within fixed environments, whereas RUSS must achieve a "perception-decision-control" closed-loop in dynamic interaction scenarios. This imposes higher requirements on the real-time performance, robustness, and flexibility of hardware-software interfaces. The modular design of RUSS often requires the integration of heterogeneous hardware from multiple sources (e.g., US machines, robotic arms, navigation devices). However, differences in communication protocols and data formats between these components significantly increase the complexity of system integration. Misalignment between the frequency of US image acquisition and transmission and the control cycle of the robotic arm may lead to control oscillations or motion overshoot. RUSS systems that rely on high-precision force/position sensors and GPUs for real-time control and imaging face issues such as system complexity, limited robustness in dynamic environments, and poor portability in clinical settings. For instance, camera-based probe localization systems can estimate depth but are susceptible to interference from surgical room occlusions and are difficult to integrate with lightweight robotic arms.

Some RUSS are designed for specific organs but focus primarily on low-level technical aspects, lacking a high-level design approach from the overall system perspective. Future research should focus on co-optimizing hardware and software at the system level: enhancing the physical compatibility between hardware and the environment on the hardware side, and designing lightweight adaptive algorithms on the software side to improve the algorithm's fault tolerance concerning hardware devices and clinical environments.

### F. Ethical Considerations in the Deployment of RUSS

The clinical deployment of RUSS presents a range of ethical challenges, particularly in the areas of patient safety, accountability in autonomous decision-making, and data privacy. Addressing these issues requires an integrated approach involving technical innovation, legal frameworks, and ethical oversight. Readers interested in standardized frameworks for ethical and

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autonomous robotic systems may refer to IEC/TR 60601-4-1 [169] and IEEE 7007 [170], which provide foundational guidance on autonomy levels and ontological principles for ethically driven AI and robotics.

As discussed in Section IV-A, patient safety remains the foremost ethical concern in RUSS applications. Although RUSS can reduce operator-related variability and human error through standardized motion paths and precise force control [169], their adaptability to complex anatomical structures and unpredictable clinical scenarios must be rigorously validated. Ensuring scanning stability across different patient populations and environments is crucial to minimizing diagnostic errors or mechanical failures. To this end, real-time fault detection algorithms should be developed, and large-scale clinical trials are needed to evaluate system safety under abnormal conditions. Systems validated only in phantom or laboratory settings require further scrutiny before clinical trust can be established [171].

Accountability becomes increasingly complex in highly automated RUSS systems integrated with AI. Currently, RUSS operating at higher levels of autonomy (LoA 2 or LoA 3) have not yet been clinically translated [20]. As autonomy increases, the risk of ambiguous liability arises. It must be clarified whether such errors stem from algorithmic flaws, operator misuse, or clinical decision-making. To address this, a clearly defined responsibility framework is necessary. Interpretable AI models should be developed to ensure transparency in the decision-making process, enhancing clinical trust and facilitating accountability [172]. Incident reporting mechanisms should also be implemented to allow healthcare professionals to report AI-related errors or adverse events promptly, fostering a culture of accountability within healthcare institutions [173].

Data privacy is another critical ethical concern, particularly in AI-integrated RUSS systems. The US images and patient health data collected by RUSS contain highly sensitive personal information, and must comply with stringent data protection regulations such as the General Data Protection Regulation (GDPR). Patients should be informed about the degree of automation and how their data will be used through a clear informed consent process [174]. Several technical safeguards can be employed to minimize privacy risks. Differential privacy can obscure individual identities by adding statistical noise to the data; de-identification techniques can remove identifiable features from US images; and end-to-end encryption with strict access controls can secure data during storage and transmission. Furthermore, federated learning [175] —a decentralized approach that trains models locally and only shares aggregated parameters—offers an alternative that avoids transferring raw patient data. However, its application in RUSS may be limited by high computational demands and communication costs, particularly in resource-constrained clinical settings.

### G. Regulatory and Reimbursement Considerations

RUSS, especially those integrating AI for imaging or decision support, face increasingly complex regulatory and reimbursement landscapes that directly impact their clinical translation and widespread adoption. Regulatory agencies such as the U.S. Food and Drug Administration (FDA) and the European Union (EU) have issued updated guidance to address emerging software-based medical technologies, including AI-powered systems like RUSS. Readers interested in the approval of intelligent medical devices in the EU and the U.S. are referred to the comparative analysis by Muehlematter et al. [176] for an in-depth understanding of relevant regulatory frameworks.

The FDA's 2023 guidance "Content of Premarket Submissions for Device Software Functions" [177] outlines a risk-based framework for classifying medical software. For RUSS integrating AI for automated imaging or decision support, the guidance requires comprehensive premarket submissions, including software verification, cybersecurity measures, and risk management in line with ISO 14971 and IEC 62304. These standards raise the regulatory bar for clinical trials, especially for high-stakes diagnostic applications.

In the EU, the Medical Device Regulation (MDR) introduces Article 117 [178] to ensure that medical devices, including AI-integrated RUSS, meet the General Safety and Performance Requirements (GSPR). For RUSS that functions as combination products or incorporates autonomous diagnostic features, manufacturers are required to obtain a notified body opinion to verify device compliance before initiating clinical trials. This added regulatory layer significantly impacts the trial approval process by necessitating more detailed preclinical documentation, rigorous assessment of software safety and performance, and early planning for post-market surveillance. As a result, the pathway to clinical translation for RUSS becomes more complex, requiring multidisciplinary coordination and increased investment to meet evolving compliance expectations.

Beyond regulatory approval, RUSS must navigate the often-overlooked but equally critical issue of reimbursement. Without recognition in standardized billing systems such as Current Procedural Terminology (CPT) codes in the U.S., RUSS procedures may not be eligible for insurance reimbursement [179]. This creates an economic disincentive for hospitals and clinicians to adopt even clinically validated systems. Therefore, manufacturers and researchers must proactively generate evidence demonstrating clinical utility, workflow efficiency, and cost-effectiveness—factors that are increasingly important in health technology assessments [180]. Regulatory compliance, ethical transparency, and reimbursement readiness together form the foundation for successful translation of RUSS into routine clinical care.

## V. Future Trends

To overcome the current limitations, future research should prioritize enhancing the generalization capabilities of RUSS for diverse clinical scenarios. In this paper, we will give examples from application scenarios, diagnosis and treatment, and technological innovations.

### A. Application Scenarios

The most commercially successful RUSS in the clinic today is the tele-operated RUSS[181], which allows doctors to control the robot remotely and view high-quality images in real-time,

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making expert diagnosis accessible to patients in remote areas [182], [183]. A study [184] has shown that since April 2003, teleoperated RUSS has been in large-scale clinical trials in France. Up to the time of writing this review, the fully autonomous RUSS has not been used in large-scale clinical applications. Although clinical US has demonstrated broad utility in a variety of situations, RUSS has yet to achieve comparable versatility and effectiveness. This section will explore various application scenarios of US technology. Some of these are emerging fields where RUSS and US technology can advance together, potentially driving innovative developments. In contrast, other areas demonstrate significant contributions of clinical US yet remain largely unexplored by RUSS technology. Understanding these contrasting scenarios can help identify opportunities for RUSS to bridge existing gaps and expand its clinical relevance.

***1) Point-of-Care US:*** Point-of-care US (POCUS) is extensively used in emergency medicine to quickly assess patients and guide immediate clinical decisions [185]. POCUS can assist in diagnosing pain, swelling, and infection. It is especially useful in emergencies where rapid diagnosis is crucial. In particular, it helps detect pleural effusions, lung consolidation, and pneumothorax [186]. The use of POCUS enables the clinician to perform the US scan either at the medical office or the patient's bedside and is used as an extension of the physical examination. Real-time images can immediately be correlated with the patient's symptoms, and any changes in a critical patient's condition can be more rapidly detected [187].

To date, The University of Rochester (URMC) [188] has deployed approximately 1000 Butterfly's POCUS across 64 academic departments and clinical programs. URMC plans to scale the program to 2500 devices by 2026. These achievements have contributed to earlier diagnoses, such as detecting cholecystitis, abscesses, and bladder masses, significantly improving patient outcomes while reducing emergency visits. Notably, this implementation has resulted in a remarkable 116% increase in hospital charges, underscoring the tangible impact of POCUS integration. POCUS has already revolutionized everyday clinical practice [189], and it is believed that RUSS with POCUS will change how US is applied in daily practice.

***2) Gynecology and Obstetrics:*** Ultrasonography has become an indispensable and preferred method of clinical examination in obstetrics and gynecology. US is widely used for prenatal assessment of fetal growth, anatomy, and management of multiple pregnancies. All pregnant women should receive routine US examinations of the fetus [190]. A minimum of three screening tests should be performed during pregnancy [191]. These exams encompass a range of objectives, including assessing fetal position, detecting structural abnormalities, and evaluating placental function. RUSS needs to obtain diagnostic US images while ensuring the safety of both the fetus and mother [192]. Due to safety considerations, the application of RUSS in gynecology and obstetrics remains highly constrained. Safety concerns primarily arise from the necessity to maintain precise control over probe pressure, contact force, and scanning trajectory, all while accommodating the unique and dynamic anatomical changes that occur throughout pregnancy. Additionally, maternal movement, variations in fetal position, and complex maternal-fetal interactions further complicate the automation of these procedures. The application of RUSS in this domain is, therefore, highly constrained, with current systems [193] primarily focused on experimental validations rather than clinical implementation. There are studies [194] on fetal US imaging that have not been deployed to robotic arms. Tsumura et al. [195] developed a hardware platform for the positioning of the US probe to obtain diagnostic US images of the fetus and pregnant woman, but it has not received widespread attention.

### *B. Diagnosis and Treatment*

The application of RUSS extends far beyond conventional US scanning, with broad prospects in both medical diagnosis and treatment. By enabling more precise and efficient automated US diagnosis and therapy, RUSS holds the potential to significantly enhance the accuracy of disease screening and early detection. In the therapeutic domain, RUSS can be widely applied to US-guided interventions, such as real-time monitoring and precise control of high-intensity focused US (HIFU) for tumor ablation. Ning et al. [196] developed an automatic and precise image-guidance system that integrates multimodal information from preoperative MRI and intraoperative real-time US images, enabling intelligent diagnosis and treatment planning for HIFU therapy. Furthermore, US is a promising minimally invasive tool for modulating central nervous system function. For instance, Angolano et al. [197] successfully applied focused US (FUS) technology to disrupt the blood-brain barrier (BBB) in patients with glioblastoma. In the future, RUSS has the potential to achieve sufficient BBB disruption without causing acute damage, thereby expanding its potential in the treatment of neurological disorders.

### *C. Technological Innovations*

Leveraging advancements in the universal foundation model [198], robot learning [199], and innovative system design [200] have the potential to enhance the intelligence, scalability, and operational efficiency of RUSS.

***1) Universal Ultrasound Foundation Model:*** The introduction of vision foundation models, such as the Segment Anything Model (SAM) [201] and SAM2 [202], have driven transformative advancements in the field of computer vision. These models are characterized by their robust feature extraction capabilities and exceptional generalization performance, opening up new possibilities for research and applications across various domains [203]. In the field of medical imaging, specialized models like MedSAM [204], [205] have emerged, demonstrating outstanding performance and adaptability in tasks such as medical image segmentation and object detection. Specifically, in the domain of US imaging, the development of models such as Universal Ultrasound Foundation Model (USFM) [206], UniUSNet [207], Ultrasound Representation Foundation Model (URFM) [208], UltraSam [209], and Multi-Organ Foundation model (MOFO) [210] has provided tailored solutions by integrating the unique characteristics of US images with the powerful generalization capabilities of foundation models.

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Applying these foundation models to RUSS has the potential to significantly enhance their intelligence and adaptability. These models not only improve the stability and robustness of RUSS in dynamic environments but also enable them to handle complex anatomical variations and patient-specific features more effectively. Such advancements bestow RUSS with enhanced generalization capabilities, accelerating their translation from laboratory research to clinical practice.

***2) Light-Weighting and Portable Systems:*** Most existing RUSS rely on industrial robots or collaborative robots. These systems, such as KUKA [211], UR [212], Franka [213], etc., offer researchers a comprehensive platform to explore control and imaging. However, the conventional robot systems and robot arms were not originally designed specifically for US imaging tasks. Industrial robots require a significant amount of space for settlement, severely limiting the portability of US imaging itself and further constraining the application scenarios of RUSS [214]. Additionally, the high cost of collaborative robot platforms makes it challenging to widely promote and apply RUSS [215]. Bao et al. [216] designed a low-cost 6-DoF RUS with a serial structure for breast imaging. Ning et al. [108] introduced a portable and lightweight RUS designed to enhance adaptability for US imaging tasks. These efforts effectively reduce costs incurred due to industrial robots. The development of portable and lightweight RUSS has expanded its potential applications in diverse scenarios, including home health management, emergency care, and battlefield medicine. These systems address the limitations of traditional industrial RUSS, such as bulkiness and high cost, by offering portability and efficiency. They provide innovative solutions for primary healthcare and home-based medical scenarios, significantly improving the accessibility and efficiency of medical resources. By utilizing lightweight materials and simplified structures, these systems achieve enhanced mobility and can be deployed in dynamic, unstructured environments, such as field settings or remote diagnostics. Future advancements in portable and lightweight RUSS will focus on enhancing adaptability, cost-effectiveness, and application diversity.

***3) Learning-Based Methods:*** RUSS can be greatly improved by incorporating advances in the field of robot learning. Robot learning refers to the process by which robots simulate human learning behaviors, improving their performance through continuous learning, thereby significantly enhancing adaptability and intelligence [217]. Learning-based methods emphasize generating actions as outputs while observing the environment as inputs. Among these, supervised learning, RL, and IL have emerged as core methodologies for addressing task autonomy, skill replication, and adaptability in clinical settings [218]. For instance, IL enables robots to process unstructured environments [81], and RL formalizes robotic behavior [219]. Advanced machine learning can understand the "language of sonography" based on a limited number of expert presentations [165]. A schematic representation of learning frameworks in RUSS is depicted in Fig. 8. Traditional robots rely on fixed algorithms or pre-defined steps to complete tasks, which is both time-consuming and inflexible [220]. In clinical applications, these systems require recalibration when patient-specific variations alter parameters and often struggle in complex, unstructured scenarios. Designing learning-based RUSS with multi-modal perception and autonomous decision-making is a promising approach. These systems can learn through environmental interaction and adapt dynamically to complex, unfamiliar scenarios, similar to how clinicians operate.

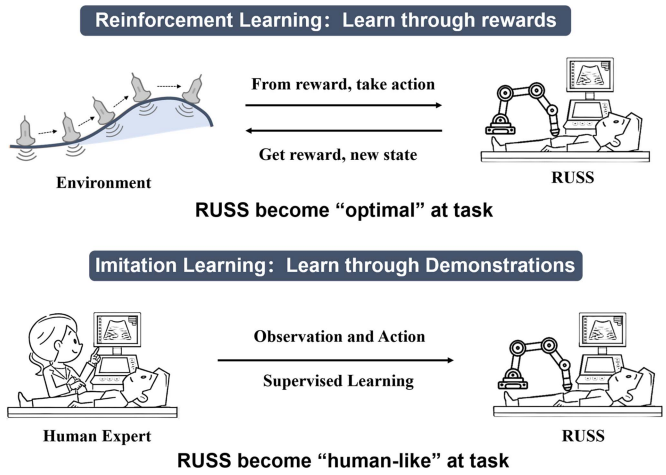



Fig. 8. Comparative framework of learning-based methods (RL and IL) in RUSS.

## VI. Discussion and Conclusion

RUSS has achieved substantial advancements in automating US scanning, minimizing reliance on operator expertise, and improving diagnostic reproducibility. Despite these benefits, significant challenges remain in clinical implementation, as the translation of RUSS from research prototypes to widespread clinical adoption is still limited. This review systematically examines the current state of RUSS, identifying the disparities between research priorities and clinical needs, as well as the barriers to real-world deployment. Key limitations include the adaptability of RUSS to clinical environments, the generalization of robotic systems across diverse anatomical structures, and their seamless integration into existing medical workflows. To address these issues comprehensively, this review is structured around an organ-specific and task-centric framework, which enables a detailed examination of RUSS technologies to both their anatomical applications and the clinical functions they aim to support.

While the long-term vision of a fully generalizable RUSS is compelling, developing a universal system in the near term poses significant technical and economic challenges, potentially leading to over-engineered solutions. A more pragmatic approach involves prioritizing specific, high-value clinical applications with moderate technical complexity and strong clinical acceptance, such as abdominal US. As a deep multi-organ imaging modality with high usage frequency and operational safety, abdominal US represents an ideal candidate for early-stage RUSS development. Modular RUSS designs tailored to distinct anatomical regions can enhance clinical adoption by addressing targeted scanning tasks efficiently. Focusing on prevalent applications with high

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patient burden offers a pathway to faster clinical acceptance compared to pursuing a fully generalizable solution.

To accelerate the clinical adoption of RUSS, we propose a clear three-phase development strategy. In the first phase, efforts should center on establishing the fundamental capabilities of RUSS, including autonomous scanning and standard plane acquisition, particularly in well-suited scenarios such as abdominal US. This phase emphasizes modular system design, real-time control, and efficient robot learning to ensure stable image acquisition and reproducibility. The second phase should advance toward more complex clinical tasks by integrating diagnostic reasoning, disease classification, and therapeutic guidance, thereby enabling RUSS to assist in decision-making and intervention. The third phase involves in vivo testing and real-world clinical validation to ensure safety, robustness, and seamless integration into existing clinical workflows. Achieving these milestones requires close interdisciplinary collaboration among engineers, clinicians, and industry partners. Through this staged, task-driven strategy, RUSS can progressively mature from experimental prototypes into clinically deployable systems—ultimately realizing the vision of translating RUSS from bench to bedside.

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